Breno Uliana de Angelo

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Research interests: mechanisms of human ageing, multi-omics, graph neural networks, knowledge graphs

Education

M.Sc. in Informatics 2026 – Jun 2027 (expected)
Universidade Federal do Espírito Santo (UFES) Vitória, Brazil

- Advisor: Alberto F. De Souza. GPA: 8.70/10 (first semester completed).
- Research: neural models for genome analysis to identify markers promoting human longevity.
- Coursework: Human Genetic Disorders, Genetic and Cellular Therapies (in progress); Bioinformatics (2027).

B.Sc. in Electrical Engineering 2021 – 2025
Universidade Federal do Espírito Santo (UFES) Vitória, Brazil

- GPA: 9.37/10. Normalized GPA (CRN): 16.84, on UFES's normalized scale (mean 5.0, one standard deviation = 1.25 points), which ranks students relative to their cohort.
- Thesis: *Evaluation of a Depth-Augmented ACT Model for Humanoid Manipulation* (advisor: Alberto F. De Souza).

Technical Diploma in Electrotechnics (high school) 2018 – 2021
Instituto Federal do Espírito Santo (IFES) Vitória, Brazil

Research & Professional Experience

Graduate Researcher (M.Sc.) 2026 – present
LCAD, UFES Vitória, Brazil

- **Genomics with AlphaGenome:** for the ICANN 2026 paper (AlphaGenome-predicted RNA-seq signals over 11 pigmentation genes, 1000 Genomes, 5 superpopulations, 2,237 training individuals), implemented the Random Forest and XGBoost baselines (PCA to 411 components, grid searches) and ran the hyperparameter search for all models. On the test set the CNN reached 84% accuracy versus 81% for XGBoost and 69% for Random Forest; a single known pigmentation gene (OCA2) gave 64% accuracy versus 37% for a random gene (FBXO5).
- **Gene prioritization (under review):** substantially improved the pipeline's code and proposed the baseline for using it to rank genes for a phenotype of interest. One classifier per gene is trained on frozen-AlphaGenome melanocyte RNA-seq predictions for 1,072 individuals; the resulting ranking recovers 9 known pigmentation genes among 33 controls at AUC 0.880 versus 0.724 for a variant-level association test on the same cohort, and 4 of its top 5 genes are known pigmentation genes (2 for the association test). In-silico perturbation experiments with AlphaGenome are in progress.
- **Software, `genomics visualize`:** built a local web app that runs the whole pipeline on a cohort: VCF import, AlphaGenome predictions for each individual, comparison of predicted functional tracks across groups, classifier training, and in-silico editing of an individual's DNA to see how the prediction changes (code; overview at breno-de-angelo.github.io).
- **Virtual Embryo Challenge:** member of the LCAD team; helped frame the problem and design the models.
- **Humanoid manipulation (Prometheus):** supervise undergraduate students continuing the project: testing different vision-language-action (VLA) models, collecting new datasets, and writing a hierarchical program for autonomously preparing a cup of coffee (in progress).

Undergraduate Research Fellow (Scientific Initiation) 2022 – 2025
LCAD, UFES Vitória, Brazil

- Autonomous Computational Intelligence project: AI for autonomous vehicles and robots (localization, mapping, perception, navigation).
- **Hercules, autonomous forklift (first author, JISEM 2025):** with colleagues, built an Ackermann robot with a manipulator arm that autonomously moves cellulose bales from a pile to a truck in a scaled-down warehouse, commanded in natural language through speech and an LLM. Programmed the arm motion; designed and integrated the bale-recognition system (YOLOv8 fine-tuned on 859 images, with depth-camera 3D localization in ROS 2); designed the low-level control system and integrated it into the autonomy stack; and contributed to the chat-based command interface. In the paper's benchmark, GPT-4 reached 100% accuracy approving feasible tasks and 83.3% in rejecting unfeasible ones; the best open-source model (Mixtral) reached 53.3% and 23.3%.
- **Argos, quadruped (Unitree Go1):** installed and tested open-source autonomy stacks.

- **Prometheus, humanoid (Unitree G1), B.Sc. thesis:** improved an open-source VR teleoperation stack (Meta Quest 3) with a new "Pinch Retargeting" hand-control strategy and stereoscopic feedback; curated 30 pick-and-place demonstrations (LeRobot format); trained and evaluated RGB-only and depth-augmented (PointNet) ACT policies.

Backend Developer, DevOps 2023 – present
AUMO S.A. Brazil

- Startup selling a SaaS and on-premise platform for chatbot applications grounded in company knowledge via retrieval-augmented generation (RAG), with connectors to Google Drive, OneDrive, Jira and others, integrations via the Model Context Protocol (MCP), and contract-based training of task-specific small language models (SLMs).
- Led the research behind a semantic routing layer that classifies chat messages (relevant, unrelated, chitchat, spam) before they reach the LLM (SBSI 2026): on a balanced 6,760-message Portuguese dataset, an embedding-similarity router reached 98.5% accuracy (BERT-based: 99.0%) versus 53.8% for zero-shot GPT, cutting cost by 96% (US$18.10 to US$0.32 per million requests) and memory from 6.5 GB to 1.0 GB.
- Responsible for Python backend development and DevOps for large-scale LLM-based NLP applications.

Volunteer, Community Outreach Project 2022 – 2024
UFES Electrical Engineering, SECRI Vitória, Brazil

- Volunteered with a UFES extension project providing technical support (electrical installations, lab setup) to a social-assistance institution for at-risk children and adolescents.

Member, PET Elétrica (Tutorial Education Program) 2022
UFES Electrical Engineering Vitória, Brazil

- Helped create AutoCAD and Python mini-courses for electrical engineering students; taught technology to older adults in the UnAPI project.

Publications

- A. F. Souza, T. Oliveira-Santos, C. Badue, **B. U. de Angelo**, D. Meira, A. Garcez, G. Guizzardi. Inferring Superpopulation Labels from DNA via AlphaGenome-Derived Functional Signals. *International Conference on Artificial Neural Networks (ICANN 2026)*, Padua (LNCS). Accepted for oral presentation; presented by me.
- **B. U. de Angelo**, G. G. Zanetti, A. F. Souza, C. Badue, A. G. Jacobsen, T. Oliveira-Santos. Text Message Routing System for Chat-Based Applications. *XXII Simpósio Brasileiro de Sistemas de Informação (SBSI)*, 2026, pp. 674–692. Accepted for oral presentation; presented by me.
- **B. U. de Angelo**, A. G. C. Pacheco, R. F. Freitas, I. J. O. Silva, E. P. De Abreu, L. F. S. C. Cardoso, T. Oliveira-Santos, C. Badue, A. F. De Souza. An Interactive Autonomous Forklift Robot Based on Large Language Models. *Journal of Information Systems Engineering and Management*, 10:946–961, 2025.

Manuscripts Under Review

- **B. U. de Angelo**, T. Oliveira-Santos, C. Badue, A. d'Avila Garcez, G. Guizzardi, A. F. De Souza. Phenotype-Associated Gene Discovery with AlphaGenome. Under review.

Skills & Languages

Programming & data: Python (pandas, NumPy), SQL.
Machine learning: PyTorch, Hugging Face Transformers, scikit-learn, XGBoost; multi-GPU training (CUDA); LLM-based applications; computer vision.
Genomics: AlphaGenome; VCF processing (bcftools, pysam); PLINK and genome-wide association testing.
Infrastructure: Docker, Kubernetes, cloud (AWS/GCP), ROS 2.
Languages: Portuguese (native), English (fluent), Spanish (good listening and reading)

Awards & Honors

- 15 medals at Brazilian national olympiads (2018–2021), including gold in Mathematics (OBMEP, OMIF), Physics (OBF), Astronomy (OBA) and Sciences (ONC), and silver in Robotics (OBR).

Thesis Committees

Committee member for two undergraduate theses on LLM-based agents for business processes and enterprise databases (UFES, advisor: Alberto F. De Souza).